

Demonstration of Lattestex

In the 13 TeV era

Number 6, Number 1

Universidade de São Paulo

9 September 2026







Some centred text

Some more centred text

Some centred text

2026-09-09

9 September 2026

September 2026

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An attached ROOT data file:

↓ data.root

- URL: <http://info.cern.ch/hypertext/WWW/TheProject.html>
- Hyperlink: TheProject
- Hyperlink: ATL-COM-PHYS-2014-1471

- “quotation” ‘marks’

- $H^+ \rightarrow tb$
- Lepton p_T and η

- Normal text: The beam energy is 6.5 TeV per beam.
- Mathematics: $E_{\text{beam}} = 6.5 \text{ TeV}$.
- With `siunitx`: The beam energy is 6.5 TeV per beam.
- Momentum: The PETRA beams at DESY were each 24 GeV/ c .
- Percentage: The selection efficiency is 95 %.

Jet pull is a variable that can be constructed from particles within a jet cone. The procedure is as follows:

- Select a pair of jets in an event.
- Build a vector sum of calorimeter cells within each jet.

$$\vec{p} = \sum_i \frac{E_T^i |r_i|}{E_T^{\text{jet}}} \vec{r}_i$$

- $\vec{r}_i$: position of jet cell i relative to jet centre
- E_T^i : transverse energy of cell i
- E_T^{jet} : transverse energy of jet

Each cell is assigned to the closer jet in (η, ϕ) space.

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Python code inline is like this: `print("hello")`

Python as a code block is like this:

```
import multiprocessing

def function(x):
    return x * x

if __name__ == '__main__':
    pool = multiprocessing.Pool(processes=4)
    result = pool.apply_async(function, [10])
    print(result.get(timeout=1))
    print(pool.map(function, range(10)))
```

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```

```
import sys
import time

def is_prime(n):
    return zip([True, False], decompose(n))[-1][0]

class IsPrimeCached(dict):
    def __missing__(self, n):
        r = is_prime(n)
        self[n] = r
        return r

is_prime_cached = IsPrimeCached()

def primes():
    yield 2
    n = 3
    while n < sys.maxint - 2:
        yield n
        n += 2
        while n < sys.maxint - 2 and not is_prime_cached[n]:
            n += 2

def decompose(n):
    for p in primes():
        if p*p > n: break
        while n % p == 0:
            yield p
            n /= p
    if n > 1:
        yield n
```

```
#include <iostream>
#include <thread>

void say_hello{};

int main(int argc, char** argv){

    std::thread my_first_thread(say_hello);
    my_first_thread.join();

    return 0;
}

void say_hello(){
    std::cout << "hello from a thread" << std::endl;
}
```

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#include <thread>

void say_hello[];

int main(int argc, char** argv){

    std::thread my_first_thread(say_hello);
    my_first_thread.join();

    return 0;
}

void say_hello[]{
    std::cout << "hello from a thread" << std::endl;
}
```

```
\documentclass{article}
\title{Processes involving charged mesons}
\author{C. ~M. ~G. ~Lattes, H. ~Muirhead, G. ~P. ~S. ~Occhialini, C. ~F. ~Powell}
\begin{document}
\maketitle
```

In recent investigations with the photographic method, it has been shown that slow charged particles of small mass, present as a component of the cosmic radiation at high altitudes, can enter nuclei and produce disintegrations with the emission of heavy particles. It is convenient to apply the term 'meson' to any particle with a mass intermediate between that of a proton and an electron.

```
\end{document}
```

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```

```
total 136
drwxrwxr-x 15 user user 4096 Sep  8 17:12 .
drwxrwxr-x  3 user user 4096 Sep  8 17:36 ..
drwxrwxr-x  2 user user 4096 Sep  8 17:12 analyses
drwxrwxr-x  2 user user 4096 Sep  8 17:12 anytree
drwxrwxr-x  2 user user 4096 Sep  8 17:12 bin
drwxrwxr-x  5 user user 4096 Sep  8 17:12 CI
-rw-rw-r--  1 user user 2445 Sep  8 17:12 CMakeLists.txt
drwxrwxr-x  7 user user 4096 Sep  8 17:12 configs
-rw-rw-r--  1 user user 1552 Sep  8 17:12 CONTRIBUTING.md
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-rwxrwxr-x  1 user user 6767 Sep  8 17:12 getSharedFiles.sh
drwxrwxr-x  8 user user 4096 Sep  8 17:12 .git
-rw-rw-r--  1 user user  125 Sep  8 17:12 .gitignore
drwxrwxr-x  3 user user 4096 Sep  8 17:12 .gitlab
-rw-rw-r--  1 user user 20268 Sep  8 17:12 .gitlab-ci.yml
-rw-rw-r--  1 user user   85 Sep  8 17:12 .gitmodules
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-rw-rw-r--  1 user user 1069 Sep  8 17:12 LICENSE
-rw-rw-r--  1 user user 2260 Sep  8 17:12 mkdocs.yml
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-rw-rw-r--  1 user user  140 Sep  8 17:12 requirements
drwxrwxr-x  2 user user 4096 Sep  8 17:12 Root
drwxrwxr-x  2 user user 4096 Sep  8 17:12 scripts
-rw-rw-r--  1 user user  752 Sep  8 17:12 setup.sh
```

Here is some terminal output:

```
drwxrwxr-x 15 user user 4096 Sep  8 17:12 .
drwxrwxr-x  3 user user 4096 Sep  8 17:36 ..
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drwxrwxr-x  2 user user 4096 Sep  8 17:12 HHARD
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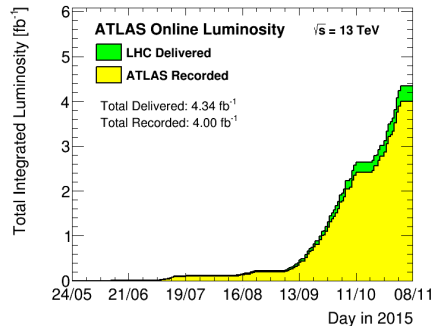
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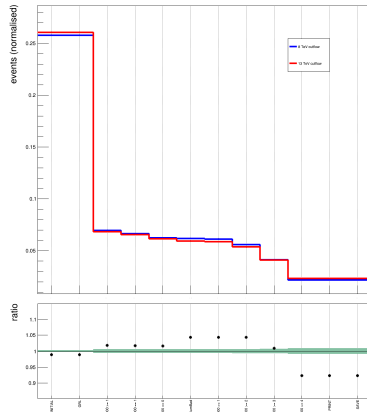
- Suppressed with respect to other Higgs modes
- $H \rightarrow b\bar{b}$ has the largest branching ratio (0.577 for m_H 125 GeV)
- Irreducible background from $t\bar{t}b\bar{b}$
- Other backgrounds: $t\bar{t}$ production in association with light quarks (u, d, s) or gluon jets (called $t\bar{t} + \text{light}$), and $t\bar{t} + c\bar{c}$



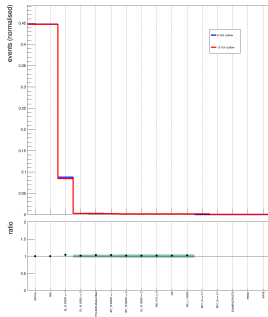
$\sqrt{s}$ [TeV]	7	8	13	14
$\sigma(t\bar{t}H)$ [pb], $m_H = 125 \text{ GeV}$	0.086	0.130	0.5085	0.611
$\sigma(t\bar{t})$ [pb]	177	253	832	950
$S/\sqrt{B}$	0.00646	0.0082	0.0176	0.0198

7 TeV $\rightarrow$ 13/14 TeV: $S/\sqrt{B}$ changes by factor of $\simeq 3$

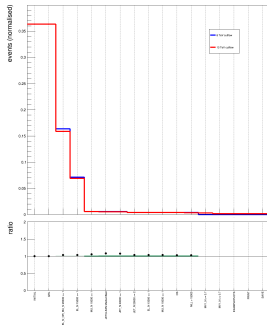
8 TeV cutflow versus 13 TeV cutflow for selection mu_jets



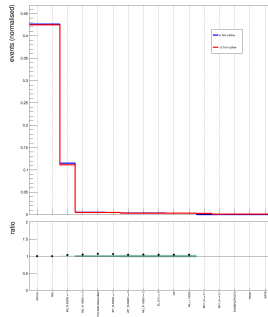
8 TeV cutflow versus 13 TeV cutflow for selection ee



8 TeV cutflow versus 13 TeV cutflow for selection emu

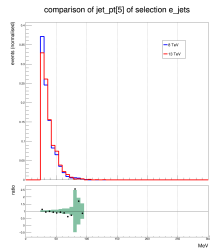
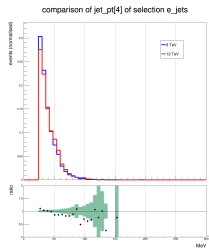
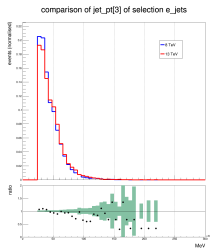
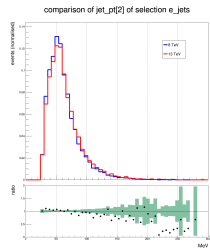
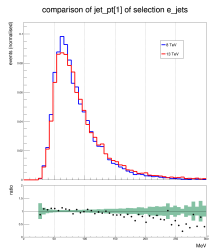
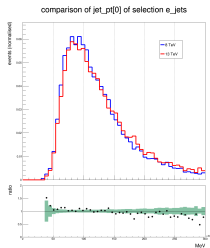


8 TeV cutflow versus 13 TeV cutflow for selection mumu

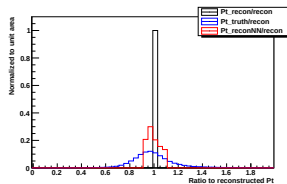


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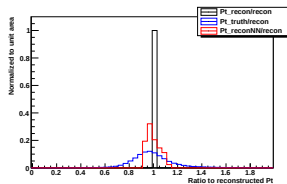
[//indico.cern.ch/event/387410/contribution/9/material/slides/0.pdf](https://indico.cern.ch/event/387410/contribution/9/material/slides/0.pdf)



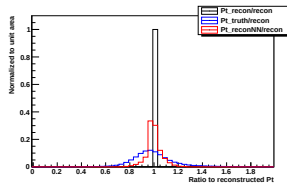
40 epochs:



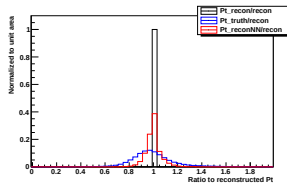
100 epochs:



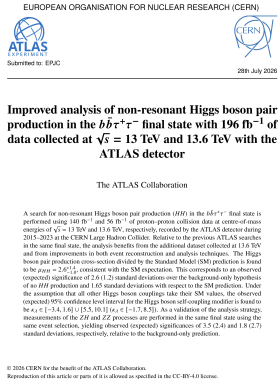
145 epochs:



300 epochs:



- Analysis paper (full Run-2 140 fb^{-1} at $\sqrt{s} = 13 \text{ TeV}$ & partial Run-3 56 fb^{-1} at $\sqrt{s} = 13.6 \text{ TeV}$) approved for ICHEP (final signoff 28 July)
- <https://gitlab.cern.ch/atlas-physics-office/HIGP/ANA-HIGP-2024-37/ANA-HIGP-2024-37-PAPER>
- <https://cds.cern.ch/record/2967281/files/HIGP-2024-37-002.pdf>
- Supporting internal note:
<https://cds.cern.ch/record/2943884>
- Main ICHEP presentation:
<https://indico.cern.ch/event/1522800/contributions/6987254>
- Preparations underway for full Run-3 analysis:
<https://indico.cern.ch/event/1713662>



$M_{b\bar{b}}$ resolutions for $VH_{b\bar{b}}$ for progressively decreasing MET energy cut requirements for various neural networks, shown to 3 significant figures:

Selection	Events	NN0	NN1	NN2	NN3
$VH_{b\bar{b}}$	23686	0.133	0.129	0.131	0.131
$VH_{b\bar{b}} + \text{MET} < 100 \text{ GeV}$	22654	0.132	0.130	0.129	0.131
$VH_{b\bar{b}} + \text{MET} < 70 \text{ GeV}$	21094	0.131	0.128	0.129	0.129
$VH_{b\bar{b}} + \text{MET} < 40 \text{ GeV}$	15050	0.128	0.126	0.126	0.126
$VH_{b\bar{b}} + \text{MET} < 20 \text{ GeV}$	6174	0.130	0.127	0.126	0.127

Here, the physical processes are ranked according to the effectiveness of the corresponding behaviour they induce in NN3, where a greater effectiveness is taken to mean a smaller resolution value. *Caveat:* Systematic uncertainties are not given their due consideration.

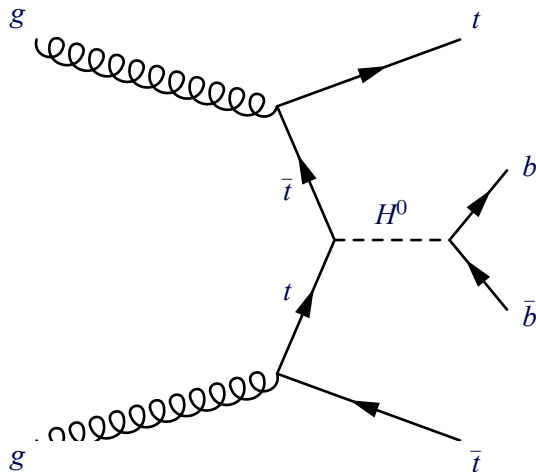
Selection	Events	NN0	NN1	NN2	NN3
$VH_{b\bar{b}} + \text{MET} > 100 \text{ GeV}$	1032	0.121542	0.129038	0.13072	0.116975
$VH_{b\bar{b}} + \text{MET} < 40 \text{ GeV}$	15050	0.128387	0.125939	0.125637	0.125963
$VH_{b\bar{b}} + \text{MET} < 20 \text{ GeV}$	6174	0.129539	0.127454	0.126029	0.127043
$VH_{b\bar{b}} + \text{MET} < 70 \text{ GeV}$	21094	0.131248	0.128119	0.128908	0.128825
$VH_{b\bar{b}} + \text{MET} < 100 \text{ GeV}$	22654	0.132004	0.129924	0.129095	0.130467
$VH_{b\bar{b}}$	23686	0.132823	0.129032	0.131202	0.131303
$VH_{b\bar{b}} + \text{MET} > 20 \text{ GeV}$	17512	0.135974	0.13137	0.13366	0.132341
$VH_{b\bar{b}} + \text{MET} > 40 \text{ GeV}$	8636	0.140116	0.135415	0.140013	0.140551
$VH_{b\bar{b}} + \text{MET} > 70 \text{ GeV}$	2592	0.143505	0.15228	0.151469	0.155914

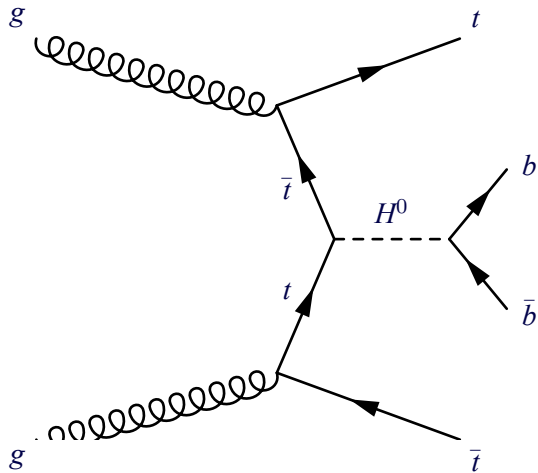
Gaussian-fit $m_{b\bar{b}}$ resolution for selected training epochs:

Subset	Epoch			
	40	100	145	300
Training	0.137	0.138	0.138	0.138
Training test	0.139	0.139	0.139	0.139

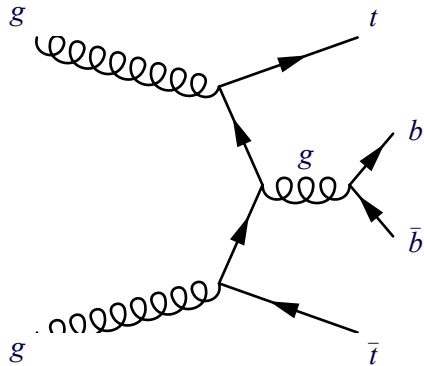
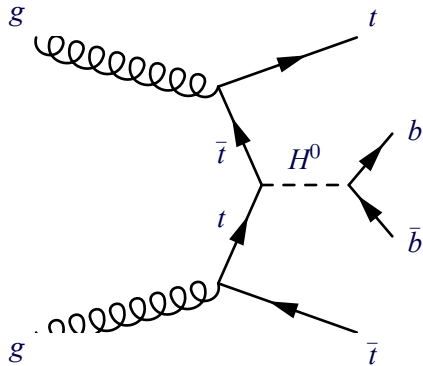
Comparison of Gaussian-fit $m_{b\bar{b}}$ resolutions with and without MET as an input variable:

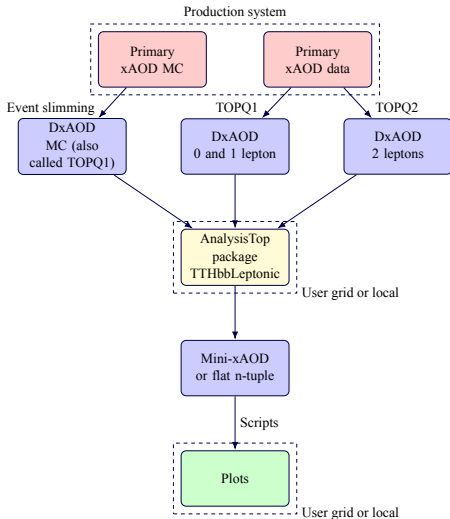
Epoch	Input	$l\nu bb$	$llbb$	$\nu\nu bb$	All
50	Without MET	0.135159	0.138616	0.135159	0.137488
	With MET	0.130047	0.137266	0.136842	0.138516
	Change	-3.78 %	-0.97 %	+1.24 %	+0.75 %
100	Without MET	0.134537	0.138781	0.136560	0.137430
	With MET	0.129719	0.137265	0.136247	0.138948
	Change	-3.58 %	-1.09 %	-0.22 %	+1.10 %
150	Without MET	0.136760	0.138464	0.137943	0.137470
	With MET	0.138292	0.137261	0.137344	0.138948
	Change	+1.12 %	-0.87 %	-0.43 %	+1.07 %
500	Without MET	0.139041	0.139451	0.138490	0.138270
	With MET	0.139225	0.137261	0.136398	0.138948
	Change	+0.13 %	+1.60 %	-1.51 %	-0.48 %





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